

Fleet Management System (FMS)

Custom Service API Tutorial Guide

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Document History

The following table shows the latest and previous versions of this document and describes the important changes.

Version	Date	Change Notice
V0.3	10.21.2021	First Creation

1. Introduction

The purpose of this document is to create a Job through the web socket API and provide a simple guide to check the Job's status and robot's status with virtual FMS server.

2. Connect to FMS Server

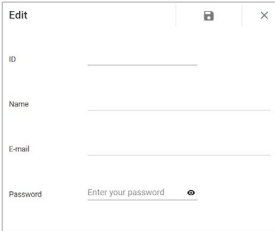
2.1. Web Socket Connection

- Address: ws://ws-open-api.robotconcert.org:20002
 - “ws-open-api.robotconcert.org” is example. It’s an ip address of the FMS server.
- Task: Gocart180 Delivery(60f6904612c350aff5366cf2)
- Preset:
 - 1F Goto Cafe-home("60f7f1998edb53002c2ef129")
 - Move to the “Café-home” location
 - 3F Goto 3rd_office_location_1("60f7f4018edb53002c2ef12b")
 - Move to the “3rd_office_location_1” location
 - 1F Cafe-Delivery & Dock("60f7f2d58edb53002c2ef12a")
 - Move to the “Cafe-home” location
 - Wait 10sec
 - Move cafe_table_1
 - Wait 10sec
 - Dock on “1f-dockingStation” station

2.2. Log In to Balcony & Share Preset

2.2.1. Create a User

- 1.Connect to the Balcony(<http://ws-open-api.robotconcert.org/>).
- 2.Login with Admin account.
- Default password: 1234
- 3.Move to User Management page. Click () on left sidebar.
- 4.Click () button.
- 5.Input ID, Name, E-mail, Password and click Save () Button.



2.2.2. Share Preset

1. Move to Preset page. Click () on left sidebar.
2. Click share icon () on the upper right side.
3. Choose a user to share the preset.

3. API Samples using Preset

3.1. Log In

Request

```
{
  "uuid": "this_is_request_id",
  "method": "req_login",
  "params": {
    "id": "admin",
    "passwd": "1234"
  }
}
```

Response

```
{
  "isSuccess": true,
  "uuid": "this_is_request_id",
  "method": "res_login"
}
```

3.2. req_get_presets

Request

```
{
  "uuid": "a4d01f42-c4d6-4fe1-8aa6-69e6b77fa7f1",
  "method": "req_get_presets",
  "params": {
    "detail ": false
  }
}
```

3.3. req_get_preset

Query a specific preset.

Request

```
{
  "uuid": "8a4f9193-cba6-4c4a-9846-4229d60f884e",
  "method": "req_get_preset",
  "params": {
    "id": "60f7f2d58edb53002c2ef12a",
    "detail": true
  }
}
```

```
}
}
```

3.4. req_exec_preset

Create an on-demand job from the preset.

Notice

1. Need to remember the “Report ID” for the corresponding id of the feedback message.
 2. When a job is successfully created, the events sent by the server begin to be received.
- Assumption: serviceId='SimpleService'
 - Because API accounts with the same “serviced” have the same pub-sub pattern, the server sent event can be received if the WebSocket connection has been established.
 - Therefore, the connection for receiving server-sent and the connection for the FMS server can be used separately. That is, the connection for monitoring and API can be used separately.

Request

```
{
  "uuid": "26e935cb-9a08-4548-8d14-58da9c2beaf2",
  "method": "req_exec_preset",
  "params": {
    "id": "60f7f2d58edb53002c2ef12a"
  }
}
```

Response

```
{
  "isSuccess": true,
  "uuid": "26e935cb-9a08-4548-8d14-58da9c2beaf2",
  "method": "res_exec_preset",
  "payload": {
    "id": "6115dc54681432002b24d78a"
  }
}
```

Feedback-report (pending)

```
{
  "uuid": "01f0f2a6-cb52-4e8a-aec2-ecf9ca2f35bb",
  "method": "feedback-report",
  "payload": {
    "id": "6115dc54681432002b24d78a",
    "title": "1F Cafe-Delivery & Dock",
    "worker": null,
    "created_at": "2021-08-13T02:43:32.975Z",
    "assigned_at": null,
    "started_at": null,
    "finished_at": null,
    "status": "pending"
  }
}
```

Feedback-report (assigned)

```
{
  "uuid": "ddd3037e-546a-4df2-a63a-f5190af3e332",
  "method": "feedback-report",
}
```

```

"payload": {
  "id": "6115dc54681432002b24d78a",
  "title": "1F Cafe-Delivery & Dock",
  "worker": "60ed00cc96f020002b9d046b",
  "created_at": "2021-08-13T02:43:32.975Z",
  "assigned_at": "2021-08-13T02:43:33.650Z",
  "started_at": null,
  "finished_at": null,
  "status": "assigned"
}
}

```

Feedback-report (started)

```

{
  "uuid": "89d526f6-b414-41d9-bf45-99eac3a6a25b",
  "method": "feedback-instruction",
  "payload": {
    "reportId": "6115dc54681432002b24d78a",
    "workerId": "60ed00cc96f020002b9d046b",
    "started_at": "2021-08-13T02:43:33.844Z",
    "finished_at": null,
    "updated_at": "2021-08-13T02:43:33.865Z",
    "status": "started",
    "id": "5405688557d3cb5606853802c8b62690"
  }
}

```

Feedback-report (running) – repeated until finished

```

{
  "uuid": "39d08640-f71c-4e88-8848-75340d085181",
  "method": "feedback-report",
  "payload": {
    "id": "6115dc54681432002b24d78a",
    "title": "1F Cafe-Delivery & Dock",
    "worker": "60ed00cc96f020002b9d046b",
    "created_at": "2021-08-13T02:43:32.975Z",
    "assigned_at": "2021-08-13T02:43:33.650Z",
    "started_at": "2021-08-13T02:43:33.844Z",
    "finished_at": null,
    "status": "running"
  }
}

```

Feedback instruction (started) – Repeat according to the number of instructions

```

{
  "uuid": "89d526f6-b414-41d9-bf45-99eac3a6a25b",
  "method": "feedback-instruction",
  "payload": {
    "reportId": "6115dc54681432002b24d78a",
    "workerId": "60ed00cc96f020002b9d046b",
    "started_at": "2021-08-13T02:43:33.844Z",
    "finished_at": null,
    "updated_at": "2021-08-13T02:43:33.865Z",
    "status": "started",
    "id": "5405688557d3cb5606853802c8b62690"
  }
}

```

Feedback instruction (done)

```

{
  "uuid": "f7658b83-793a-41c4-b02b-c1295bffc29a",

```



```

"method": "feedback-instruction",
"payload": {
  "reportId": "6115dc54681432002b24d78a",
  "workerId": "60ed00cc96f020002b9d046b",
  "started_at": "2021-08-13T02:43:51.436Z",
  "finished_at": null,
  "updated_at": "2021-08-13T02:43:51.445Z",
  "status": "started",
  "id": "9c34813ee4f09fe10a3a8054e24524b6"
}
}

```

Feedback report (finished)

```

{
  "uuid": "f7658b83-793a-41c4-b02b-c1295bffc29a",
  "method": "feedback-instruction",
  "payload": {
    "reportId": "6115dc54681432002b24d78a",
    "workerId": "60ed00cc96f020002b9d046b",
    "started_at": "2021-08-13T02:43:51.436Z",
    "finished_at": null,
    "updated_at": "2021-08-13T02:43:51.445Z",
    "status": "started",
    "id": "9c34813ee4f09fe10a3a8054e24524b6"
  }
}

```

3.5. req_get_report

Query a specific report (a record of job performance).

Request

```

{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_report",
  "params": {
    "id": "6115dc54681432002b24d78a",
    "detail": false
  }
}

```

Response

```

{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_get_report",
  "payload": {
    "id": "6115dc54681432002b24d78a",
    "title": "1F Cafe-Delivery & Dock",
    "worker": "60ed00cc96f020002b9d046b",
    "status": "finished",
    "created_at": "2021-08-13T02:43:32.975Z",
    "assigned_at": "2021-08-13T02:43:33.650Z",
    "started_at": "2021-08-13T02:43:33.844Z",
    "finished_at": "2021-08-13T02:44:49.620Z",
    "updated_at": "2021-08-13T02:44:49.629Z"
  }
}

```

3.6. req_get_reports

Query the list of reports on the specific date (UTC+0).

Notice

- Assuming: You do not use the Restful API at all.

If the real-time response is not processed, it is intended to synchronize data in either long-polling or batch format, but the size of the report is not small and should not be used as much as possible.

Request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_reports",
  "params": {
    "yyyymmdd": "2021-07-21"
  }
}
```

3.7. req_get_remain_reports

Query a list of reports that have not yet been completed. It should be used if short polling is used.

Request

```
{
  "uuid": "6232c16e-98ad-477b-83a1-acb777d385dd",
  "method": "req_get_remain_reports"
}
```

3.8. req_cancel_report

Cancel a running or pending report by the user. The report will be returned to the payload.

Request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_cancel_report",
  "params": {
    "id": "60f7fd96f020002b9d0546",
    "message": "cancel from ws-open-api",
    "detail": true
  }
}
```

Response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_cancel_report",
  "payload": {
    "id": "60f7d5fa6cea710041ad9f61",
    "title": "New Job : 2021-07-21 5:08:16 pm",
    "worker": "60c02561d5bccb002b9c68b4",
    "status": "finished",
    "created_at": "2021-07-21T08:08:26.451Z",
  }
}
```

```

    "assigned_at": "2021-07-21T08:08:27.402Z",
    "started_at": "2021-07-21T08:08:27.571Z",
    "finished_at": "2021-07-21T08:09:07.711Z",
    "updated_at": "2021-07-21T08:09:07.712Z"
  }
}

```

4. Job Compose Sample

4.1. req_get_task

Query a specific task with name

- i.e., Gocart180 Delivery(60f6904612c350aff5366cf2)

Request

```

{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_get_task",
  "params": {
    "name": "Gocart180 Delivery"
  }
}

```

4.2. req_get_task_worker

Query Worker that can perform Task

- Gocart180 Delivery(60f6904612c350aff5366cf2).

Request

```

{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_get_task_worker",
  "params": {
    "name": "Gocart180 Delivery"
  }
}

```

4.3. req_compose_job

Create a job by Task ("Gocart180 Delivery").

- The ID of the instruction should be given to be distinguishable.
- The ID will be an ID of "Server-Sent-Event".

Actions

	Instruction	Request payload
1	Move (cafe_table_5:"5f1a3f11cc6e2a00120d17db")	{ "func_name": "move", "args": {"destination": "5f1a3f11cc6e2a00120d17db"}}

2	Wait 10sec	{ "func_name": "wait", "args": {"type": "duration", "timeout": 10000}}
3	Dock (1f-dockingStation: "60f7f266a71141002ac5afd4")	{ "func_name": "dock", "args": {"station": "60f7f266a71141002ac5afd4"}}

Note: The “destination” parameter of the “Move” instruction should be the “id” of the resource. You can get the location information from below URL.
(open API doc: <http://15.165.165.134:10605/api-docs/#!/Resource>)
15.165.165.134:10605/api/resources?resource_type=Location

```
{
  "description": "cafe_table_5",
  "resource_active": true,
  "resource_type": "Location",
  "_id": "5f1a3f11cc6e2a00120d17db",
  "map": "5f19503fdbb58b001973eca1",
  "pose": {
    "x": 2.3296100591030147,
    "y": -0.7513268623612035,
    "theta": 4.67536886451076
  },
  "type": "waypoint",
  "name": "cafe_table_5",
  "created_at": "2021-07-13T02:27:57.976Z",
  "updated_at": "2021-07-13T02:27:57.976Z",
  "id": "5f1a3f11cc6e2a00120d17db"
},
```

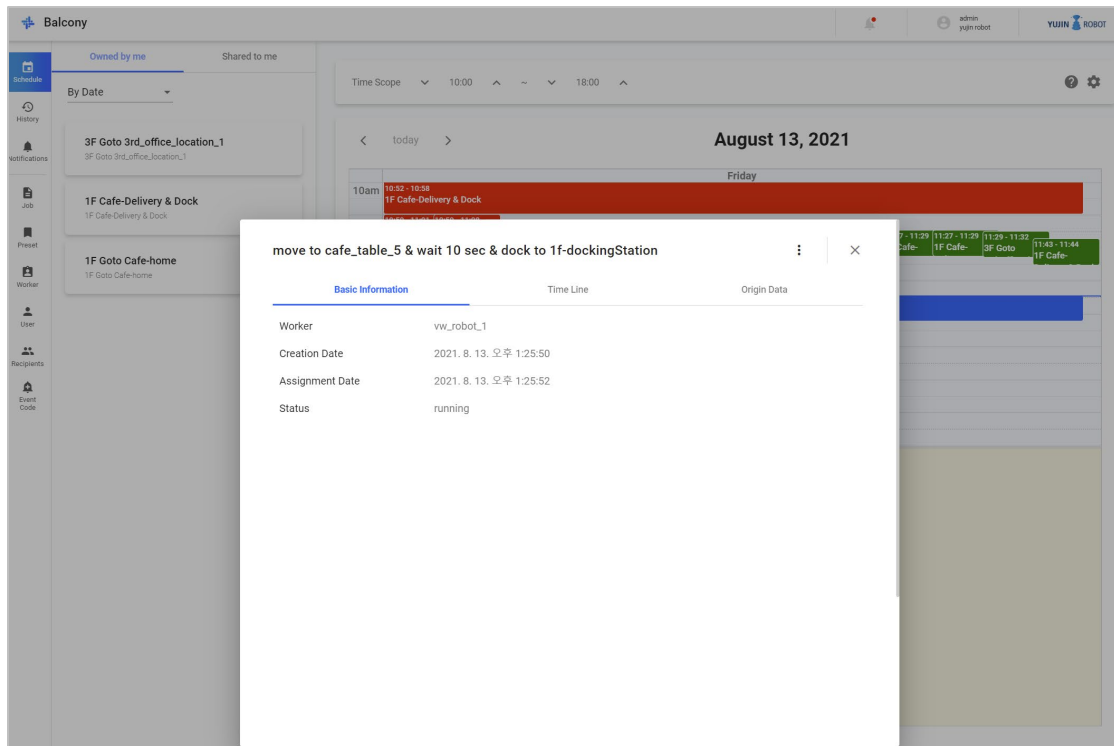
The “station” parameter of the “Dock” instruction should be the “id” of the resource. You can get the station resource information from below URL.
http://15.165.165.134:10605/api/resources?resource_type=ChargingStation

Request

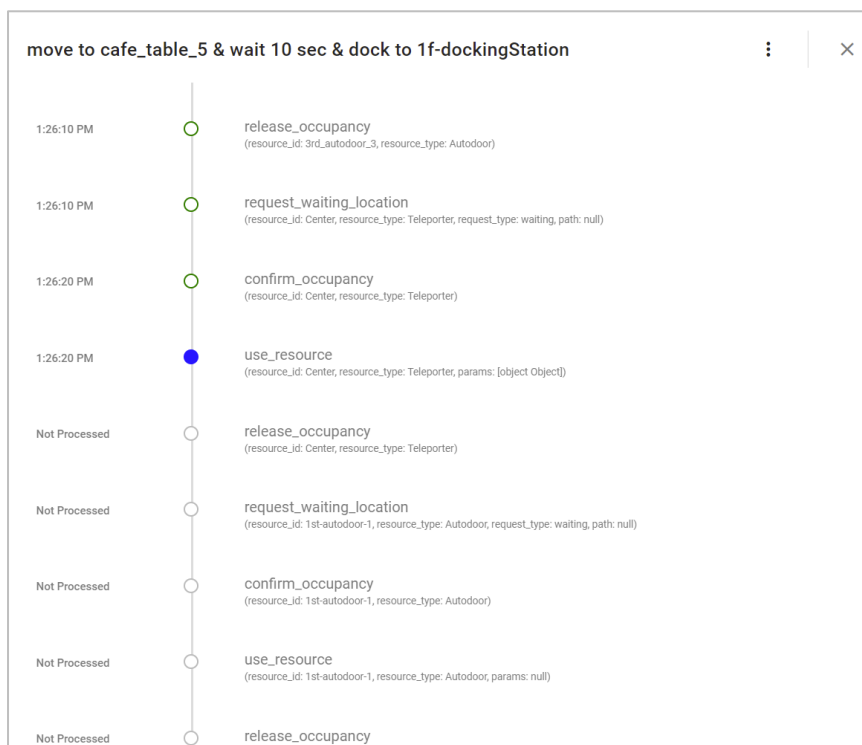
```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_compose_job",
  "params": {
    "title": "move to cafe_table_5 & wait 10 sec & dock to 1f-dockingStation",
    "task": "Gocart180 Delivery",
    "instructions": [
      { "id": "1", "func_name": "move", "args": {"destination": "5f1a3f11cc6e2a00120d17db"}},
      { "id": "2", "func_name": "wait", "args": {"type": "duration", "timeout": 10000}},
      { "id": "3", "func_name": "dock", "args": {"station": "60f7f266a71141002ac5afd4"}}
    ]
  }
}
```

Result

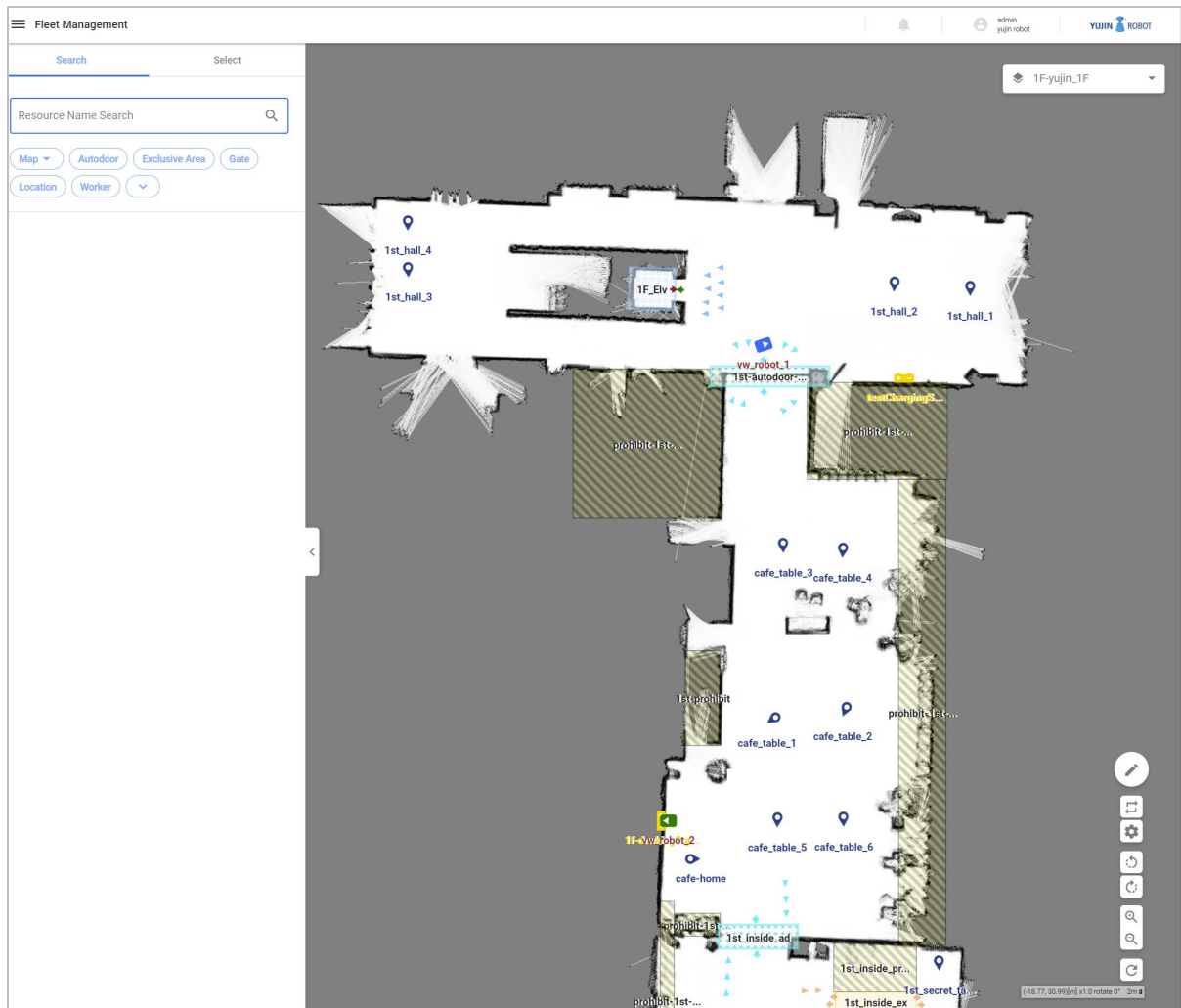
- Balcony (<http://15.165.165.134/schedule>)



- Timeline



- FMUI (<http://15.165.165.134:10901/monitoring>)

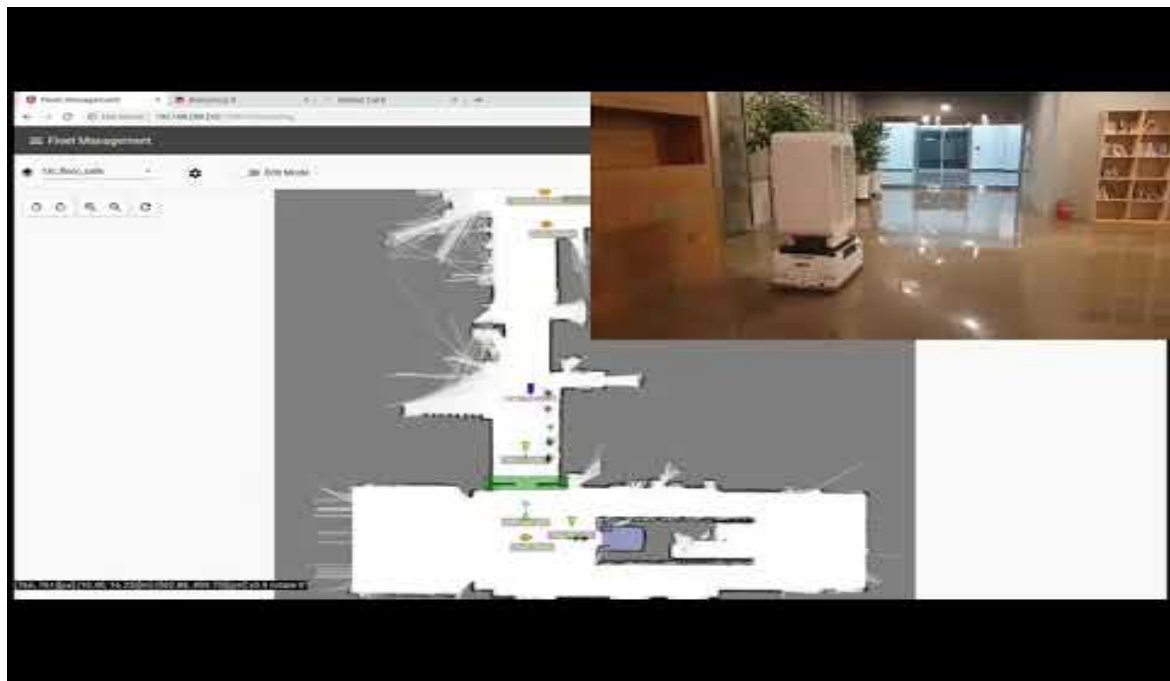


5. Useful Tool

[Web socket king client](#)

6. Reference Movie Clip

[AMS Use case: Cafe delivery in YUJIN ROBOT building](#)



The design and specifications of the products may change without prior notification for quality improvement.



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